

Gil Jorge

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Portugal
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EXPERIENCE AND EDUCATION

Aqua Hotels, Sales/Marketing, Lagoa— Paid *Internship*

July 2022

Developed automated accounting data analysis workflow producing daily dashboards. Managed website's chatbot (client service) and reservations.

Champalimaud Center for the Unknown, Lisbon— *Internship*

Jan 2026-present

6 month internship at Paton's Learning lab. Currently working on the project:

Clockless Alignment: Developing an Algorithm to Infer Neural-Behavior Correspondence. where I temporally align electrophysiological data through hardware artifacts and biological patterns with state machine data from the same experiment. Studying the intersection of natural/artificial intelligence.

Instituto Superior Técnico, Lisbon— *Bsc Physics Engineering*

Sep 2023 - July 2026 GPA out of 20: Year 1- 16; Year 2- 15; Year 3- 17

Completing **210 credits** (ECTS) instead of the usual 180 by taking extra courses (Abstract Algebra, Statistical Physics, Knowledge Representation and Reasoning of Msc CS, Cryptography and Security Protocols and Mathematics for Machine Learning of Msc Applied Math). Other courses: Deep learning of Msc CS 17/20, Discoveries in Modern Physics, 19/20 for researching and presenting on Quantum Information and Computation, and a project at the Quantum Photonics Lab with 19/20. **One of 7 students** of my cohort to earn a spot at oral exam in calculus (went in multivariable calculus, calculus 2).

SKILLS (software + languages)

Python (numpy, matplotlib, pytorch);
C++ (root), **R**, **Latex**, **HTML**, **JS**, **CSS**,
Emacs Org mode (Lisp), **SQL**, **Matlab**
Linux- Lubuntu in a 2012 laptop, maximizing its efficiency software wise for its hardware, now became a self hosted server on Nextcloud. Using Mint on new laptop.
Adobe Premiere; **Davinci Resolve**
Portuguese; C1 English; Advanced Spanish; Intermediate French.

HONORS

- Featured on Portimão Jornal for **maximum grades** (20/20) in all courses in high school.
- Level 5 music formal education (theory+guitar+chorus)
- Discover EU ambassador
- Part of the class featured in national journals for the making of [national gardens day](#)

PROJECTS

Writing (giljorge0.github.io):

300 thousand written words, on [personal website](#) preparing selected pieces for publication on public platforms. Mostly philosophy.

Multisport athlete (<https://www.youtube.com/@giljorge7479>):

Compete in university's national track and field championships, represented Sporting Clube de Portugal for 2 years. Also competing in Greco-Roman/freestyle wrestling, recently obtaining 4th place at national championships, and now train in Portugal's high performance training center with the national team. Non competitively practicing calisthenics/streetslifting, olympic weightlifting, powerlifting. Highlights of the last ones include human flag, one arm pull up and are available [online](#) along with video editing projects.

Digital brain (These and other coding projects available on github.com/giljorge0):

Built a local-first neuro-symbolic memory system integrating symbolic knowledge graphs, vector retrieval, and temporal reasoning for LLM-assisted querying. Optimized a NetworkX/SQLite architecture to scale beyond 50k nodes using NumPy batched operations and deployed a FastAPI + Ollama backend with LangGraph orchestration. Achieved a 14% MRR improvement over baseline semantic RAG and developed continual-learning mechanisms using Louvain clustering and PageRank-based knowledge-gap detection.

Emergent AI-AI communication and geometry of meaning research programme:

Designed 23 experiments across emergent communication under channel constraints (discreteness as error detection, ZK causal metrics and failure-mode taxonomy, opacity sweeps, 4-knob unified grid search, Shannon Pareto frontier) and multilingual concept geometry (LaBSE centroid gap scores, translation validity, dense AI-AI protocol beating text at k=4, Universal Grammar probes at scale, frame role induction, phoneme mapping, language-design loss). Confirmed MI entanglement between agent world models and showed message length dominates joint constraint optimisation.

